

INC-2026-0519 - Inventory Synchronisation Drift and Oversells

Incident Report | Engineering

Document ID	INC-2026-0519	Seller ID	S002
Department	Engineering	Classification	CONFIDENTIAL
Owner	Vikram Singh	Created Date	2026-05-22
Related Scenario ID	SC-002	Document Type	Incident Report

1. Incident Summary

Field	Value
Incident ID	INC-2026-0519
Severity	SEV-2
Service	Inventory Sync Service
Component	Nightly stock batch pipeline
Detected	2026-05-19 09:10 IST
Mitigated	2026-05-19 10:35 IST (seller disabled listings)
Resolved	2026-05-21 11:30 IST
Incident owner	Vikram Singh, Engineering
Seller affected	S002 - BluePeak Electronics
Related ticket	TKT-2026-0519-207

Published stock levels for seller S002 drifted up to 26 hours behind the seller's actual warehouse position, against a six-hour freshness requirement. 18,400 SKUs carried stale availability and 412 orders were accepted for items that were out of stock.

Two independent faults were required to produce the drift: a scheduler applying a fixed UTC offset where IST was expected, and a stale SKU mapping table left in the pipeline after a catalogue restructure.

2. Impact

Dimension	Detail
SKUs with stale availability	18,400 of approximately 142,000
Maximum observed staleness	26 hours
Oversell orders	412
Customer impact	412 orders cancelled and refunded
Seller impact	Affected listings disabled for approximately 29 hours
Duration of drift	Undetermined start; confirmed from at least 17 May
Data loss	None

The undetermined start date is itself a finding. There was no freshness metric, so the earliest point at which drift can be evidenced is the earliest retained batch log, 17 May. The drift may have begun earlier.

3. Detection

Detected by the seller. BluePeak reported at 08:15 IST on 19 May that it was receiving orders for items its warehouses showed as zero stock, and supplied 40 example order identifiers.

Support confirmed 26-hour staleness at 09:10 and escalated at 09:30.

The batch job had reported success on every run. Monitoring tracked whether the job completed, not whether the data it published was current. A job that succeeds while publishing day-old data is indistinguishable, in that monitoring, from a healthy job.

4. Timeline

Time (IST)	Event
17 May 02:00	Earliest batch run evidenced as publishing stale data
19 May 08:15	Seller raises ticket with 40 example oversell orders
19 May 08:52	Seller escalates by phone as oversells continue
19 May 09:10	Support confirms 26-hour data staleness
19 May 09:30	Escalated to Engineering and Operations
19 May 10:15	Seller agrees to disable affected listings
19 May 10:35	Listings disabled; new oversells stop
19 May 13:40	Scheduler timezone offset identified
19 May 15:10	Timezone alone found insufficient to explain 26 hours
19 May 16:20	Stale SKU mapping table identified as the second factor
20 May 09:00	Scheduler pinned to Asia/Kolkata; stale mapping removed
20 May 14:00	Verification run confirms correct window selection
21 May 11:30	SKU mapping rebuilt; full resynchronisation executed
21 May 15:00	Seller re-enables listings after verification
23 May 12:10	412 oversold orders refunded with priority handling

5. Symptoms

- Listings showed availability for SKUs with zero warehouse stock.
- Batch job logs reported success with no errors or warnings.
- Affected SKUs were distributed across both fulfilment centres but not

uniformly: the Chennai centre showed a higher proportion.

- Some SKUs were entirely absent from the published position despite being present in the source file.

- Stock for SKUs created since the March catalogue restructure was correct.

The last two symptoms were what separated this from a simple scheduling problem. A pure timezone offset would have made all data uniformly stale. The uneven distribution and the missing SKUs indicated a second

fault.

6. Affected Services

Service	Role	State
Inventory Sync Service	Consumed stock file and published availability	Operating on stale input
Batch scheduler	Selected the stock window	Defective: fixed UTC offset
SKU mapping resolver	Mapped seller SKUs to catalogue identifiers	Defective: stale table
Catalog Service	Held current catalogue version	Correct throughout
Order Service	Accepted orders against published availability	Functioning as designed

7. Investigation

7.1 The Scheduler Fault

The batch selects a stock window relative to the run time. The window calculation used a fixed UTC offset constant rather than resolving Asia/Kolkata.

BluePeak delivers its stock file at approximately 23:00 IST. The job runs at 02:00 IST. Under correct behaviour the job selects the window ending at the most recent file delivery. Under the fixed-offset calculation the window boundary fell 5 hours 30 minutes earlier, before the file was delivered, so the job selected the **previous night's** file.

This alone produced roughly 24 hours of staleness.

7.2 The Mapping Fault

The March 2026 catalogue restructure introduced new catalogue identifiers. A migration mapped old seller SKUs to new identifiers and wrote a new mapping table. The legacy table was retained as a fallback for rows that failed to resolve against the new table, and the fallback was never removed.

Rows failing to resolve against the current mapping silently resolved against the legacy table. Where the legacy mapping pointed at a catalogue identifier that had since been reassigned, stock was published against the wrong SKU. Where the legacy mapping pointed at an identifier that no longer existed, the row was dropped without report.

This explains the uneven distribution and the missing SKUs, and added up to a further two hours of apparent staleness where stock landed on the wrong identifier.

7.3 Why Neither Alone Was Sufficient

The freshness requirement is six hours. The scheduler fault alone produced about 24 hours of staleness, which would have breached the requirement had anything been measuring it. Nothing was. The mapping fault alone produced incorrect rather than stale data, which would have surfaced as individual SKU complaints rather than as a systemic pattern.

Together they produced a large, visible, systemic failure that the seller detected within days. Paradoxically, the combination is what made the problem detectable at all.

8. Root Cause

The nightly stock batch published stale and partially misdirected inventory because the scheduler computed its window using a fixed UTC offset rather than the Asia/Kolkata timezone, causing it to process the previous

night's file, and because a legacy SKU mapping table retained as a silent fallback resolved rows that should have been rejected.

9. Contributing Factors

- 1. **No freshness metric.** The pipeline measured completion, not currency.
- 2. **A silent fallback.** The legacy mapping made a failure look like a success. A strict rejection would have surfaced the problem in March.
- 3. **Implicit timezone handling.** The scheduler relied on a constant rather than an explicit timezone, a pattern not specific to this job.
- 4. **Migration cleanup incomplete.** The March restructure left a fallback in place with no removal date or owner.
- 5. **Requirement not enforced.** The six-hour freshness requirement existed in the integration specification but had no corresponding control.

10. Mitigation

The effective mitigation was the seller disabling affected listings at 10:35 on 19 May, which stopped new oversells within about 20 minutes. This was the seller's decision, taken on a call with Support, and it was the right one: continuing to trade on unreliable availability would have compounded the harm for the 24 hours needed to correct the pipeline.

No platform-side mitigation was available that would have been faster. The correct stock position could not be published until the pipeline was fixed.

11. Resolution

Action	Completed
Scheduler pinned to Asia/Kolkata explicitly	2026-05-20 09:00
Legacy SKU mapping table removed from the pipeline	2026-05-20 09:00
Verification run confirming correct window selection	2026-05-20 14:00
SKU mapping rebuilt against current catalogue version	2026-05-21 11:30
Full stock resynchronisation	2026-05-21 11:30
Seller re-enabled listings after verification	2026-05-21 15:00
412 oversold orders refunded	2026-05-23 12:10

12. Validation

Check	Method	Result
Window selection correct	Run timestamps compared against file delivery times over 7 days	Correct file selected on every run

Check	Method	Result
Published stock matches source	Full position compared against seller export	Matched within 4 minutes of run completion
No silent row drops	Rejection report reviewed	1,940 rows rejected and reported, previously silent
Freshness within requirement	Freshness metric observed for 14 days	Maximum 2 hours 10 minutes
Oversell rate	Monitored for 30 days	7 in June, all concurrent-purchase races

The 1,940 rejected rows were initially alarming. Review with BluePeak established that all were discontinued SKUs the seller had removed from its catalogue but was still including in its stock export. The seller corrected its export on 19 June and rejections fell below 40 per day.

13. Preventive Actions

ID	Action	Owner	Target	Status
PA-1	Explicit timezone for the stock batch scheduler	Vikram Singh	2026-05-20	Complete
PA-2	Remove legacy mapping fallback	Vikram Singh	2026-05-20	Complete
PA-3	Per-feed freshness metric	Neha Gupta	2026-05-29	Complete
PA-4	Alert on freshness breach above 6 hours	Neha Gupta	2026-06-05	Complete
PA-5	Unknown-SKU rows rejected and reported, never mapped	Vikram Singh	2026-06-12	Complete
PA-6	Rejection report delivered within 30 minutes of processing	Vikram Singh	2026-06-12	Complete
PA-7	Timezone audit across all scheduled jobs	Neha Gupta	2026-07-24	Complete

PA-7 found three further jobs resolving timezone implicitly. None were producing incorrect behaviour, but all three carried the same latent fault and were corrected.

14. Follow-Up

Requirements STK-2, STK-3 and STK-4 in the S002 integration specification were tightened on 8 June 2026 to require explicit timezone handling, to state the freshness ceiling as a testable requirement, and to require mapping resolution against the current catalogue version only.

Post-incident review held 22 May 2026. The review's main conclusion was that the pipeline had two silent failure modes and that silence, rather than either fault, was the real defect.